

Saturated fat intake and insulin resistance in men with coronary artery disease. The Stanford Coronary Risk Intervention Project Investigators and ...

BACKGROUND: To determine whether there is an association between diet and plasma insulin concentration that is independent of obesity, we studied the relation of dietary composition and caloric intake to obesity and plasma insulin concentrations in 215 nondiabetic men aged 32-74 years with angiographically proven coronary artery disease. **METHODS AND RESULTS:** After adjusting for age, the intake of saturated fatty acids and cholesterol were positively correlated (p less than 0.05) with body mass index ($r = 0.18$, $r = 0.16$), waist-to-hip circumference ratio ($r = 0.21$, $r = 0.22$), and fasting insulin ($r = 0.26$, $r = 0.23$). Carbohydrate intake was negatively correlated with body mass index ($r = -0.21$), waist-to-hip ratio ($r = -0.21$), and fasting insulin ($r = -0.16$). Intake of monounsaturated fatty acids did not correlate significantly with body mass index or waist-to-hip circumference ratio but did correlate positively with fasting insulin ($r = 0.24$). Intake of dietary calories was negatively correlated with body mass index ($r = -0.15$). In multivariate analysis, intake of saturated fatty acids was significantly related to elevated fasting insulin concentration independently of body mass index. **CONCLUSIONS:** These cross-sectional findings in nondiabetic men with coronary artery disease suggest that increased consumption of saturated fatty acids is associated independently with higher fasting insulin concentrations.